

Here are 5 HIGH and 3 MEDIUM questions based on the transcript:

HIGH

1. [HIG] What is Python, and why is it considered a high-level language?
2. [HIG] What are the main differences between Python 2 and Python 3?
3. [HIG] What is an expression in programming, and how does it produce a value?
4. [HIG] What is the purpose of the `terminal` or `command prompt` in Python?
5. [HIG] What is the difference between a code editor and an Integrated Development Environment (IDE)?

MEDIUM

1. [MEDIUM] How can you verify that Python is successfully installed on your system?
2. [MEDIUM] What are some of the benefits of using Python for AI, machine learning, web development, and automation?
3. [MEDIUM] What is a syntax error in programming, and how does it differ from a logical error?

Here are 8 predicted exam questions with their respective probabilities:

[HIGH] What is an expression in Python?

[MEDIUM] How does a linter work in Python?

[HIGH] What is the difference between a syntax error and a runtime error in Python?

[MEDIUM] How can you use the Len function to get the length of a string in Python?

[HIGH] What are the four best practices for naming variables in Python as mentioned in the transcript?

[MEDIUM] How do you use formatted strings to build a string with dynamic values in Python?

[HIGH] What is the purpose of using an escape character in Python strings, and how can you use it to include special characters like double quotes or new lines?

[MEDIUM] How can you use the upper method to convert a string to uppercase in Python?

Here are the predicted exam questions with their corresponding tags:

[HIGH] What is the difference between the `find()` and `in` operators when checking for the existence of a character or sequence of characters in a string?

[MEDIUM] How do you use the `strip()` method to remove leading and trailing whitespace from a string, and what are the differences between `lstrip()`, `rstrip()`, and `strip()`?

[HIGH] What is the purpose of the `replace()` method, and how can it be used to replace all occurrences of a character or sequence of characters in a string with another value?

[MEDIUM] How do you use the `round()` function to round a number to the nearest integer, and what are the differences between rounding up and down?

[HIGH] What is the difference between an `int` and a `float` in Python, and how can they be used in arithmetic operations?

[MEDIUM] How do you use the `math` module to perform mathematical calculations, such as finding the ceiling of a number using `math.ceil()`?

[HIGH] What are the differences between the `/` and `//` operators when performing division in Python, and what is the purpose of the `modulus` operator (`%`)?

[HIGH] How do you use augmented assignment operators to simplify arithmetic operations, such as incrementing a variable by a certain value?

This is a long text about Python programming, specifically covering loops (for loops, while loops) and functions. Here's a summary of the key points:

Loops

- * For loops: used to iterate over a sequence (e.g., range, list, string)
 - + Example: ``for x in range(5): print(x)``
 - + Can use ``range`` function with start, stop, and step arguments
- * While loops: used to repeat a block of code as long as a condition is true
 - + Example: ``while number > 0: print(number); number //= 2``
- * Break statement: used to exit a loop prematurely
- * Continue statement: used to skip the rest of the current iteration and move on to the next one

Functions

- * Defining a function: use ``def`` keyword followed by function name, parameters in parentheses, and indentation for code block
 - + Example: ``def greet(name): print(f"Hi {name}")``
- * Calling a function: use function name with arguments in parentheses
 - + Example: ``greet("John")``
- * Function parameters: can be required or optional (default values)
 - + Example: ``def greet(name, age=30): ...``

Range Function

- * Returns an iterable object that generates numbers from start to stop (inclusive) with a specified step size
 - + Example: ``range(1, 10)`` generates numbers 1 through 9

Infinite Loops

- * Can be created using a while loop with a condition that is always true
 - + Example: ``while True: ...``
- * Use break statement to exit the loop

The text also includes some examples and exercises to practice writing Python code.

Here are 5 HIGH and 3 MEDIUM questions based on the transcript:

[HIGH] What is the default return value of a function in Python?

[MEDIUM] How can we use keyword arguments to make code more readable when calling a function with multiple arguments?

[HIGH] What is the difference between a list and a tuple in Python?

[MEDIUM] Can you explain how to create a function that takes a variable number of arguments using an asterisk (*)?

[HIGH] Why does indentation matter in Python, especially when it comes to return statements within loops?

[MEDIUM] How can we use augmented assignment operators to simplify code, such as in the example of calculating the product of all numbers in a tuple?